

MOD 307 Course Outline

MOD 307 — Trade Policy and Gravity Modelling

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	33 (indicative band 30–36)
Shape	4 parts · 12 sessions · 8 laboratories · 4 assignments · 1 examination · 1 final project
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	Regression to the level of fixed effects and clustered standard errors. MOD 303 is recommended for the general-equilibrium context, and MOD 304 for the habit of reading a parameter file before a result. No prior gravity experience is assumed.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

The gravity equation is the most reliable empirical regularity in international economics and the easiest to estimate badly. This course estimates it properly — PPML, fixed effects, zero flows, clustered inference — and then reads the estimation this school’s own toolchain produced, which fails in three ways that a coefficient table conceals and a regressor summary reveals in one line. It goes on to the accessibility side, where costs come from engineering defaults rather than estimates, and to the policy comparison, where the flagship scenario produces an uplift of exactly zero.

Three engines were asked for a tariff-policy result and all three declined: a dropped coefficient, welfare effects of order three parts in a hundred million of a per cent, and an error message. The course reports all three rather than the one that came closest, because the pattern is the finding. Two of its registered gaps were found by tabulating a regressor before reading its coefficient — a constant column standing in for a logistics index, and an intercept wearing a fixed effect’s name. All 51 registered findings are in the appendix.

The rule the course is built on

A result that is too good is a diagnostic, not a finding. A coefficient of the expected sign and an implausible tightness usually means the regressor is not what its column name says, and

the check costs one line of code. Tabulate every regressor before you read any coefficient;
the null test is the same discipline applied to the simulation.

Who it is for

Trade economists and negotiators' analysts who estimate or consume gravity results, and modellers joining a trade-policy team who need to know which published elasticities can be reused and which cannot.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	The gravity equation and its estimation	01–04	01–02	9.0
P2	Reading a regressor before a coefficient	05–06	03–05	7.5
P3	Frictions, corridors and policy simulation	07–09	06–07	7.5
P4	Residuals, engine failure and reporting	10–12	08	6.0
	Assignment and assessment time			3.0
	Total	12	8	33.0

The 12 sessions and 8 laboratories, in teaching order, and the subject matter of each:

P1 The gravity equation and its estimation — Sessions 01–04 · 9.0 h · Laboratory 01–02

- **01 The Gravity Equation** — The empirical regularity, and how old it is; Structural derivations: Anderson–van Wincoop, Eaton–Kortum, Melitz; Multilateral resistance, and why it matters; What the equation is and is not evidence for; The canonical coefficient values; What this course's toolchain estimates
- **02 Estimation** — Zero flows: why they are informative and common; Jensen's inequality and the log of a mean; Silva and Tenreyro, and heteroskedasticity; The Poisson family without count data; Convergence and separation in PPML; Scaling the dependent variable, and what it does
- **03 Fixed Effects** — Origin and destination fixed effects; Multilateral resistance, absorbed; What is no longer identified once they are in; Pair fixed effects and time variation; Collinearity, and what gets dropped; Reading a specification for what it removed
- **04 Inference** — Clustering, and on what dimension; Why unclustered errors are almost always too small; Reading a t-statistic for plausibility; What a t of ten thousand tells you; Reporting inference honestly; The inference in this course's own estimation

- *Lab 01 Read the Extraction First* — What did this toolchain produce, and what did it fail to?
- *Lab 02 The Gravity Specification* — What was estimated, on what, with what absorbed?

P2 Reading a regressor before a coefficient — Sessions 05–06 · 7.5 h · Laboratory 03–05

- **05 Reading a Regressor Before a Coefficient** — Tabulate every regressor’s distinct values; A constant regressor and what it becomes; Near-constant regressors and weak identification; Dummies with almost no variation; What `dropcollinear` decides, and how it decides it; Why this comes before the coefficient table
- **06 An Intercept in Disguise** — The logistics variable in this panel; One distinct value across 46 countries; What collinearity dropped, and what it left; Recovering the implied intercept; The misreading the table invites; What may and may not be reported from this estimation
- *Lab 03 Tabulate Every Regressor* — Does anything in this regression have no variation?
- *Lab 04 Recover the Intercept* — What is the logistics coefficient actually measuring?
- *Lab 05 Read the Coefficients Against the Literature* — Which coefficients are plausible?

P3 Frictions, corridors and policy simulation — Sessions 07–09 · 7.5 h · Laboratory 06–07

- **07 Frictions and Accessibility** — Delivered cost and door-to-door time; The least-cost route over a corridor graph; The frictions defaults, and their roundness; Ad valorem equivalents; What the extraction says about its own numbers; Estimated frictions against engineered ones
- **08 Corridors, Borders and Nodes** — Border delays by crossing; Corridor loads and reliability; Node betweenness and bottleneck scores; Ports against land borders; One-stop border posts, and what they change; Reading a bottleneck table for policy
- **09 Policy Simulation, and the Null Test** — The seven standard policies; The null test: move a lever and check something changed; Two scenarios identical to baseline, to the last digit; Why the tariff channel is inert; Three engines, three zeros, one cause; What may be reported from the five that do move
- *Lab 06 The Null Test* — Does every policy lever actually do something?
- *Lab 07 Frictions and Bottlenecks* — Where is the cost and where is the delay?

P4 Residuals, engine failure and reporting — Sessions 10–12 · 6.0 h · Laboratory 08

- **10 Missing Trade and Trade Potential** — Actual against predicted, and the residual; Trade potential as a normative claim; The fill ratio; Why a residual is not an opportunity; How missing-trade numbers are misused; Reporting a potential estimate honestly
- **11 When Engines Fail** — Five engines, two errors; A keyword mismatch, read carefully; A precompilation failure, read carefully; What an extraction that records errors buys you; What is lost when an engine does not run; Reporting a partial toolchain
- **12 Reporting a Trade Analysis** — The three categories every number falls into; The trade note template; How gravity results are oversold; The honest statement for this toolchain; What would change it; The standard the project is marked against
- *Lab 08 The Trade Note* — What can honestly be said about this toolchain?

What a graduate can do

1. State the gravity equation and its structural derivations.
2. Explain why PPML rather than log-OLS, and what zero flows do.
3. Specify origin and destination fixed effects and say what they absorb.
4. Report inference correctly, including why clustering matters.
5. Diagnose a coefficient table by tabulating its regressors first.
6. Recognise a constant regressor absorbing an intercept.

7. Read a coefficient against the literature's range and against its sign.
8. Distinguish an estimated friction from an engineering default.
9. Run a null test on a policy lever before reporting any scenario.
10. Compute and interpret a missing-trade or trade-potential measure.
11. Read a corridor and border-delay dataset for bottlenecks.
12. Write a trade note that separates what was estimated from what was assumed and what did not run.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 96 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	35.04 %	8 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	The extraction inventory; Tabulate before you read; The null test; The trade note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete review of a trade analysis — this toolchain, or one the learner is responsible for — delivered as a nine-section trade note plus the diagnostic scripts that produced it

The per-laboratory weights are an equal split of the laboratory component rounded to two decimals, so the column sums to 100.04 rather than 100. The component weights are the binding ones.

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted data. The bilateral trade and friction data are published aggregates and ship with the course. The logistics variable in the extraction is a constant and is used only in the laboratory that exposes it; no substitute is fetched, because substituting real data would repair somebody else's extraction and hide what the course is teaching. The aggregate residual is reported as a residual and appears only in the chapter that explains why it must not be called potential.

Materials and set-up

A 120-page Technical Manual, 13 lecture decks of 134 slides, 8 laboratory decks of 102 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 96-item quiz bank, 5 assessment files, 20 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

The estimation laboratories run in R and Python and need no model instantiated. R on this host is installed but not on PATH: call Rscript by its full path, and repeat a failed library() once before believing it, for the reason the school's R note gives.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in COURSE_VERSION.json.

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